Yuxi Li

bachelor, ANU(The Australian National University), Canberra, Australia
bachelor, SDU(Shan Dong University), Shandong, China master,UM(University of Macau)
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RESEARCH INTERESTS

Research interest 1 Diophantine Geometry(especially,unlikely intersection)
Research interest 2 Arakelov geometry
Research interest 3 (other)Algebraic geometry
Research interest 4 Algebra and Category theory
Research interest 5 Analysis

EDUCATION

ANU(The Australian National University), Canberra, Australia — July 2024 — June 2025
Bachelor of Science in Mathematics — Cumulative GPA:6.5/7.0
(Credits transferred from Shandong University are not counted in the grades)

SDU(Shan Dong University , Shan dong China — Sep. 2021 — June 2025
Bachelor of Science: Mathematics — Cumulative GPA: 84.56/100 or 3.46/5.00

UM(University of Macau), Macau China — Aug. 2025 — May 2027
Master of Science: Mathematics — Cumulative GPA: Null

SELECTED COURSES(Bachelor)

(SDU) Courses

- Partial Differential Equation Modelling (ANU)
- Partial Differential Equations
- Abstract Algebra
- Supplementary Course of Ordinary Differential Equations
- Advanced Analysis I(ANU)
- Topology
- Function of Complex Variable
- Classical Mechanics I(ANU)
- Classical Mechanics2(ANU)
- Real Function Theory
- Probability Theory
- Ordinary Differential Equations
- (remark:the course with (ANU) is taken in China bu the teacher is from ANU,there are courses corresponding to them in ANU)

(ANU) Courses

- Special Topics in Maths(reading course using the GTM82,called Differential forms and Algebraic Topology)
- Stoch Analysis with Fin Appl(master)
- Advanced Functional Analysis(master)
- Algebraic Topology(master)
- Galois Theory(master)
- Differential Geometry
- Special Topics in Maths(reading course called Microlocal analysis)(master)
- Fractual Geometry(master)

SELECTED COURSES(Master)

(UM) Courses

- Real Analysis
- Advanced Probability theory

AWARDS

Mathematics Competition for (Chinese) College students
The 14th National Second prize in Mathematics Competition for College students (Mathematics Class A). — Jan. 2023

Mathematics Competition for (Chinese) College students
The 13th national college students First prize in the school competition (Mathematics A). Dec. 2021

Director's Scholarship for Academic Excellence
Director's Scholarship for Academic Excellence(in ANU,for SDU student who go to ANU with the highest GPA among the students who take part in the SDU-ANU project). May 2024

The knowledge I learnt by myself

I myself have carefully studied the knowledge:

- basic Category theory.
- basic Commutative Algebra and Noncommutative Algebra.
- Advanced Real analysis ,Advanced Complex analysis
- Elementary Number theory And (some) Algebraic Number Theory.
- Basic Set theory.
- Advanced general Topology theory.
- Classical Harmonic Analysis.

Something I've coarsely taught myself:

- some physics,such as Analytical mechanics, Quantum mechanics,
  QFT(including The CFT studied by physicists) ,
  Statistical mechanics, Electrodynamics.
- Algebraic Geometry.
- Modular form.
- Lie algebra.
- Riemann surface.
- Different kinds of Dynamic system.
- Mathematical logic.
- Differrential topology.
- Mathematics related to cybernetics and Financial mathematics.
- Cybernetics theory.
- Advanced Functinal analysis(TVS and Banach algebra)

The following is what I am learning(by my self or with course I am taking):

- Complex Geometry
- Algebraic Geometry
- Algebraic Number theory
- Diophantion Analysis

ENGLISH & GRE TESTS

IELTS (Academic):6.0

Listening: 6.0 — Reading: 6.5
Speaking: 4.5 — Writing: 6.0
Test date: 17/Jan/2025

REMARK:I have studied in ANU for a year and I will study in University of Macau for two years(English teaching)

SKILLS

- **Programming:** Having learnt C++,Python(a little).
- **Software:** Knowing how to use the Latex and Matlab.
- **Soft Skills:** learnt a little Japanese,knowing how to cook.
- **Sports:** Exercise and relax with arcade music games
- **Like subjects other than math:** Physics,Cybercnetics theory,Economics.

REFERENCES

Prof. Anand Deopurkar
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Scholar Profiles: ANU Mathematical Sciences Institute- Personal Page — Personal Page — ORCID profile

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